

Twitter Sentiment Analysis and Brand Intelligence Platform

Project Overview

Project Title

Twitter Sentiment Analysis and Brand Intelligence Platform using Machine Learning, Deep Learning, SQL, and Power BI

Objective

The objective of this project was to analyze customer opinions from Twitter data and classify sentiment using Machine Learning and Deep Learning techniques. The project also aimed to compare model performance and present business insights through interactive Power BI dashboards.

Business Goals

Goal	Description
Sentiment Analysis	Classify tweets as Positive or Negative
Brand Intelligence	Understand customer perception from social media data
Model Comparison	Compare traditional ML models with BERT
Data Analytics	Generate business insights using SQL and Power BI
Visualization	Build interactive dashboards for decision-making

Dataset Description

Dataset Information

The project uses the Sentiment140 Twitter dataset.

Column	Description
Target	Sentiment label (0 = Negative, 4 = Positive)
Ids	Unique Tweet ID
Date	Tweet timestamp
Flag	Query information
User	Twitter username
Text	Tweet content

Dataset Size

Metric	Value
Total Tweets	1.6 Million
Sentiment Classes	Positive and Negative
Source	Twitter Sentiment140 Dataset

Data Preparation and Preprocessing

The dataset was cleaned and transformed before model development.

Data Cleaning Activities

Activity	Purpose
Missing Value Validation	Verify dataset completeness
Duplicate Analysis	Identify repeated records
Text Cleaning	Remove URLs, punctuation, and symbols
Lowercasing	Standardize text format
Tokenization	Split text into tokens
Stopword Removal	Remove non-informative words
Text Normalization	Prepare text for NLP modeling

Data Storage Optimization

Process	Description
CSV Import	Raw dataset ingestion
Parquet Conversion	Improved storage and loading speed
Pandas Processing	Data analysis and transformation

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand sentiment distribution and user activity patterns.

Analyses Performed

Analysis	Purpose
Sentiment Distribution	Measure positive and negative tweets
User Activity Analysis	Identify highly active users

Analysis	Purpose
Tweet Volume Analysis	Evaluate dataset scale
Text Exploration	Understand tweet content characteristics

Sentiment Encoding

Value Sentiment

0	Negative
4	Positive

SQL Database Integration and Query Analysis

SQLite was integrated into the notebook to demonstrate structured data storage and querying capabilities.

SQL Workflow

Step	Description
Database Creation	Create SQLite database
Data Storage	Store cleaned dataframe into SQL table
Query Execution	Perform analytical SQL queries
Reporting	Retrieve business insights

SQL Queries Performed

Query	Purpose
Total Tweets	Determine dataset size
Sentiment Distribution	Count tweets by sentiment
Distinct Users	Count unique users
Top 10 Users	Identify most active users
Average Tweets per User	Measure engagement level

Business Value of SQL

- Structured data storage
 - Fast querying and reporting
 - Dashboard integration support
 - Business analytics capability
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Feature Engineering

Feature engineering transformed textual data into machine learning input features.

Techniques Applied

Technique	Purpose
TF-IDF Vectorization	Convert text into numerical features
Sentiment Encoding	Prepare target variable
Feature Selection	Improve model performance

Machine Learning Models

Multiple classification models were trained and evaluated.

Models Implemented

MODEL	CATEGORY
LOGISTIC REGRESSION	Supervised Learning
RANDOM FOREST	Ensemble Learning
XGBOOST	Gradient Boosting
BERT (DISTILBERT)	Deep Learning NLP

Model Performance Comparison

Evaluation Metrics

Metric	Purpose
Accuracy	Overall prediction correctness
Precision	Positive prediction quality
Recall	Positive detection effectiveness
F1 Score	Balance between precision and recall

Model Results

Rank	Model	Accuracy	F1 Score
1	BERT (DistilBERT)	80.30%	80.29%
2	Logistic Regression	77.10%	77.08%
3	XGBoost	75.82%	75.73%
4	Random Forest	70.94%	70.66%

Best Performing Model

Metric	Value
Model	BERT (DistilBERT)
Accuracy	80.30%
F1 Score	80.29%

Key Observation

BERT achieved the highest performance by leveraging transformer-based language understanding, outperforming all traditional machine learning models.

Project Architecture

The project followed a complete end-to-end AI analytics pipeline.

Architecture Workflow

CSV to Parquet	Storage optimization			
Notebook Loading	Pandas-based processing			
Data Validation	Check schema and integrity			
Text Cleaning	NLP preparation			
NLP Preprocessing	Tokenization and normalization			
Feature Engineering	Create model inputs			
Train-Test Split	Dataset partitioning			
ML Modeling	Train classification models			
Model Evaluation	Measure performance			
SQL Integration	Store and query structured data			
Power BI Reporting	Dashboard development			
Documentation	Record methodology and results			
Final Delivery	Project submission package			

Architecture Diagram

Figure 1: Twitter Sentiment Analysis Data Pipeline Architecture

(Insert Architecture Diagram Here)

Architecture Benefits

- End-to-end workflow visibility

- Reproducible data pipeline
- Better project organization
- Integration of AI, SQL, and BI tools

Power BI Dashboard Development

Four dashboards were developed to visualize sentiment insights and model performance.

Dashboard Summary

Dashboard	Purpose
Executive Overview	Dataset KPIs and sentiment distribution
User Analysis	Top users and user activity analysis
Tweet Explorer	Tweet-level exploration and filtering
Model Performance	Compare machine learning results

Dashboard Components

- KPI Cards
- Donut Charts
- Bar Charts
- Tables
- Interactive Filters
- Model Comparison Visualizations

Key Findings

Finding	Description
Large Dataset	Successfully processed 1.6 million tweets
Sentiment Classification	Effective identification of positive and negative sentiment
User Insights	Active users generated high tweet volumes
Model Comparison	BERT achieved highest performance
Dashboard Reporting	Interactive business insights were generated

Conclusion

This project successfully developed a Twitter Sentiment Analysis and Brand Intelligence Platform using Machine Learning, Deep Learning, SQL, and Power BI. The dataset was cleaned,

analyzed, and transformed into machine learning features before training Logistic Regression, Random Forest, XGBoost, and BERT models. SQL integration enabled structured querying and reporting, while Power BI dashboards provided interactive business intelligence capabilities.

Among all evaluated models, BERT (DistilBERT) achieved the highest performance with 80.30% Accuracy and 80.29% F1 Score, making it the most effective model for sentiment classification within this project.

Future Scope

- Real-time sentiment monitoring
- LSTM implementation
- SHAP explainability analysis
- Cloud deployment
- Automated reporting pipelines
- Multi-platform social media analytics